Practical: Creating a Single Page Website Using React

Aim

To create a Single Page Website using React concepts such as **Hooks, State, Props, and Router**.

STEP 1: Check Node.js Installation

Open **Command Prompt** or the **VS Code Terminal**.

Run:

node --version

Then:

npm --version

If both commands show version numbers, Node.js and npm are installed.

Example:

v22.x.x
10.x.x

STEP 2: Create a React Project

Open VS Code.

Go to:

Terminal → New Terminal

Run:

npm create vite@latest react-single-page

You will get some options.

Select:

Project name: react-single-page

Framework: React

Variant: JavaScript

STEP 3: Go Inside the Project

Run:

cd react-single-page

Now we are inside our React project.

STEP 4: Install React Packages

Run:

npm install

This installs all the packages required by the React project.

STEP 5: Install React Router

We need Router for navigation.

Run:

npm install react-router-dom

What is React Router?

React Router allows us to navigate between different views/pages of a React application.

For our practical:

Home
About

Contact

will be connected using Router.

STEP 6: Open the Project in VS Code

If the project is not already open, run:

code .

Your project will look approximately like:

```
react-single-page
│
├── node_modules
├── public
│
├── src
│   ├── assets
│   ├── App.jsx
│   ├── App.css
│   ├── index.css
│   └── main.jsx
│
├── package.json
└── index.html
```

For this practical, we mainly need:

```
src
 └── App.jsx
```

STEP 7: First Understand State

Before making the complete website, let's understand **State**.

Open:

src → App.jsx

Delete the existing code.

Write:

```
import { useState } from "react";

function App() {

  const [count, setCount] = useState(0);

  return (
    <div>

      <h1>Counter</h1>

      <h2>Count: {count}</h2>

      <button onClick={() => setCount(count + 1)}>
        Increase
      </button>

    </div>
  );
}

export default App;
```

Save the file.

STEP 8: Run the Project

In the terminal:

npm run dev

You will get something like:

Local: http://localhost:5173/

Open that URL in your browser.

You will see:

Counter

Count: 0

[Increase]

Click **Increase**.

You will get:

Count: 1

Click again:

Count: 2

STEP 9: Understand State

Look at:

const [count, setCount] = useState(0);

There are three things here:

count

This stores the current value.

setCount

This changes the value.

0

This is the initial value.

So:

Initial
count = 0

↓

Click button

↓

setCount(count + 1)

↓

count = 1

What is State?

State is data that can change during the execution of a React component.

STEP 10: Understand Hook

We wrote:

import { useState } from "react";

and:

useState(0)

`useState()` is a **React Hook**.

What is a Hook?

A **Hook is a special React function that allows functional components to use React features.**

For this practical, we use:

useState()

to manage State.

Therefore:

Hook
↓
useState()
↓
State
↓
State changes
↓
UI updates

STEP 11: Understand `onClick`

We have:

<button onClick={() => setCount(count + 1)}>

`onClick` is used to detect a button click.

When the user clicks:

Button Click
↓
setCount()
↓
count increases
↓
Screen updates

STEP 12: Now Create a Props Example

Now we will understand **Props**.

Change `App.jsx` temporarily to:

```
function Welcome(props) {

  return (
    <h2>
      Welcome, {props.name}!
    </h2>
  );
}

function App() {

  return (
    <div>

      <h1>My React Website</h1>

      <Welcome name="Student" />

    </div>
  );
}

export default App;
```

Save the file.

The browser shows:

My React Website

Welcome, Student!

STEP 13: Understand Props

We created:

function Welcome(props)

The component receives `props`.

Then:

<Welcome name="Student" />

passes:

name = Student

The Welcome component receives it through:

props.name

So:

App Component
↓
name="Student"
↓
Welcome Component
↓
props.name
↓
Welcome, Student!

What are Props?

Props are used to pass data from one component to another component.

STEP 14: Now Create the Home Component

Now we will create our actual website.

Replace `App.jsx` with the following code.

First create the Home component:

```
function Home() {

  return (
    <div>

      <h1>Home Page</h1>

      <p>
        Welcome to our React Single Page Website.
      </p>

    </div>
  );
}
```

Here:

function Home()

is a **React Component**.

STEP 15: Create the About Component

Below the Home component, write:

```
function About() {

  return (
    <div>

      <h1>About Page</h1>

      <p>
        This website is created using React.
      </p>

    </div>
  );
}
```

Now we have:

Home Component
About Component

STEP 16: Create the Contact Component

Add:

```
function Contact() {

  return (
    <div>

      <h1>Contact Page</h1>

      <p>
        Email: student@example.com
      </p>

    </div>
  );
}
```

Now:

Home
About
Contact

are our three components.

STEP 17: Add State to Home

Now we will combine **State + Hook** with our Home component.

Change the Home component to:

```
function Home() {

  const [count, setCount] = useState(0);

  return (
```

```
    <div>

      <h1>Home Page</h1>

      <p>
        Welcome to our React Single Page Website.
      </p>

      <h2>Counter: {count}</h2>

      <button onClick={() => setCount(count + 1)}>
        Increase
      </button>

    </div>
  );
}
```

Remember that we need:

```
import { useState } from "react";
```

at the top.

STEP 18: Create Props Component

Now create:

```
function Welcome(props) {

  return (
    <h2>
      Welcome, {props.name}!
    </h2>
  );
}
```

We will pass the name later.

STEP 19: Import Router

At the very top of `App.jsx`, write:

```
import {
  BrowserRouter,
  Routes,
  Route,
  Link
} from "react-router-dom";

import { useState } from "react";
```

What are these?

BrowserRouter

BrowserRouter

Enables routing in our React application.

Routes

Routes

Contains all routes.

Route

Route

Defines which component should be displayed for a particular URL.

Link

Link

Creates navigation links.

STEP 20: Create the Main App

Now create:

```
function App() {

  return (

    <BrowserRouter>

      <nav>
```

```
        <Link to="/">Home</Link>

        {" | "}

        <Link to="/about">About</Link>

        {" | "}

        <Link to="/contact">Contact</Link>

      </nav>

      <Welcome name="Student" />

      <Routes>

        <Route
          path="/"
          element={<Home />}
        />

        <Route
          path="/about"
          element={<About />}
        />

        <Route
          path="/contact"
          element={<Contact />}
        />

      </Routes>

    </BrowserRouter>
  );
}
```

STEP 21: Complete `App.jsx`

Now your complete file should be:

```
import {
  BrowserRouter,
  Routes,
```

```
  Route,
  Link
} from "react-router-dom";

import { useState } from "react";


function Home() {

  const [count, setCount] = useState(0);

  return (
    <div>

      <h1>Home Page</h1>

      <p>
        Welcome to our React Single Page Website.
      </p>

      <h2>Counter: {count}</h2>

      <button onClick={() => setCount(count + 1)}>
        Increase
      </button>

    </div>
  );
}


function About() {

  return (
    <div>

      <h1>About Page</h1>

      <p>
        This website is created using React.
      </p>

    </div>
  );
}


function Contact() {
```

```
  return (
    <div>

      <h1>Contact Page</h1>

      <p>
        Email: student@example.com
      </p>

    </div>
  );
}


function Welcome(props) {

  return (
    <h2>
      Welcome, {props.name}!
    </h2>
  );
}


function App() {

  return (

    <BrowserRouter>

      <nav>

        <Link to="/">Home</Link>

        {" | "}

        <Link to="/about">About</Link>

        {" | "}

        <Link to="/contact">Contact</Link>

      </nav>

      <Welcome name="Student" />

      <Routes>
```

```
        <Route
          path="/"
          element={<Home />}
        />

        <Route
          path="/about"
          element={<About />}
        />

        <Route
          path="/contact"
          element={<Contact />}
        />

      </Routes>

    </BrowserRouter>
  );
}


export default App;
```

STEP 22: Save and Run

Save:

Ctrl + S

If the development server is already running, the browser will automatically update.

Otherwise:

npm run dev

Open:

http://localhost:5173/

STEP 23: Test Home Page

You should see:

Home | About | Contact

Welcome, Student!

Home Page

Welcome to our React Single Page Website.

Counter: 0

[Increase]

Click **Increase**.

Counter: 1

Click again:

Counter: 2

Here we used:

useState()
↓
State
↓
setCount()
↓
UI updates

So this demonstrates **Hook + State**.

STEP 24: Test Props

At the top of the page:

Welcome, Student!

This comes from:

<Welcome name="Student" />

and:

props.name

So this demonstrates **Props**.

STEP 25: Test Router

Click:

About

The URL changes to:

/about

The screen displays:

About Page

This website is created using React.

Now click:

Contact

The URL becomes:

/contact

Output:

Contact Page

Email: student@example.com

This demonstrates **Router**.

STEP 26: Understand the Complete Working

When the website starts:

App
↓
BrowserRouter
↓
Navigation

↓
Routes

When Home is selected:

/
↓
Home Component
↓
useState()
↓
Counter

When About is selected:

/about
↓
About Component

When Contact is selected:

/contact
↓
Contact Component

Props work separately:

App
↓
<Welcome name="Student" />
↓
Welcome Component
↓
props.name
↓
Welcome, Student!

STEP 27: Final Concept Summary

1. Component

A reusable building block of a React application.

Example:

```
function Home() {
  return <h1>Home Page</h1>;
```

}

2. State

Stores data that can change.

const [count, setCount] = useState(0);

3. Hook

A special React function.

Example:

useState()

4. Props

Used to pass data between components.

<Welcome name="Student" />

5. Router

Used to navigate between different views/pages.

<Route path="/about" element={<About />} />

6. Link

Used to create navigation.

<Link to="/about">About</Link>

Final Output

React Single Page Website

Home | About | Contact

Welcome, Student!

Home Page

Welcome to our React Single Page Website.

Counter: 2

[Increase]

When **About** is clicked:

About Page

This website is created using React.

When **Contact** is clicked:

Contact Page

Email: student@example.com

Result

Thus, a Single Page Website was successfully created using React with Hooks, State, Props, and Router.